video 59 ①

Date:

How the backpropagation works in RNN:-

→ Backpropagation in RNN

RNN intro → RNN → Practical → Types of RNN

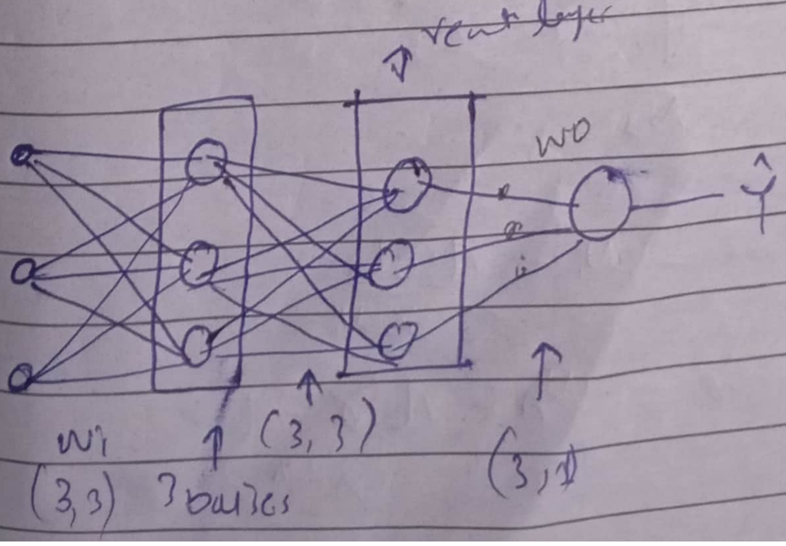
→ RNN → Back Propagation

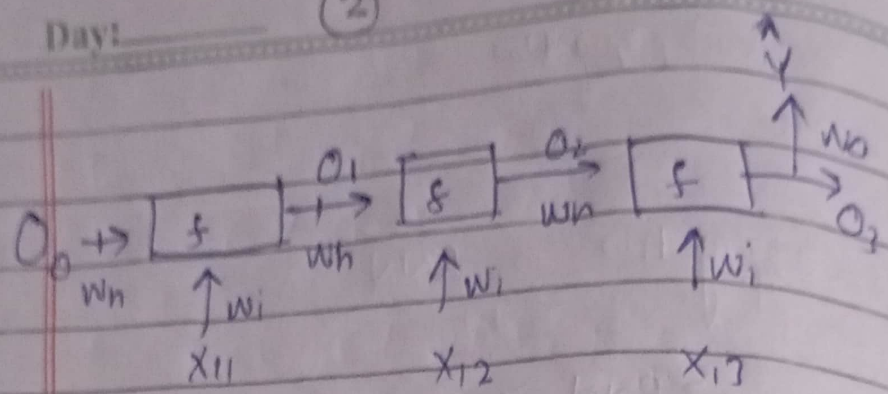
Text

cat	mat	rat	1	$X_1 \rightarrow$ [100, 010, 001]
rat	rat	rat	1	$X_2 \rightarrow$ [001, 001, 010]
mat	mat	cat	0	$X_3 \rightarrow$ [010, 010, 100]

vocabulary:-

Cat	Mat	Rat
[100]	[010]	[001]





$O_1 = f(X_{11} W_i + O_2 W_h)$

$O_2 = f(X_{12} W_i + O_1 W_h)$

$O_3 = f(X_{13} W_i + O_2 W_h)$

$Y \rightarrow \hat{y} = \sigma(O_3 W_o)$

$L = -Y_i \log \hat{Y}_i - (1 - Y_i) \log (1 - \hat{Y}_i)$

Loss calculate

↳ Minimize

L_{min} gradient descent

W_i W_h W_o

$W_i = W_i - \eta \frac{\partial L}{\partial W_i}$

$W_h = W_h - \eta \frac{\partial L}{\partial W_h}$

$$w_0 = w_0 - \eta \frac{\delta L}{\delta w_0}$$

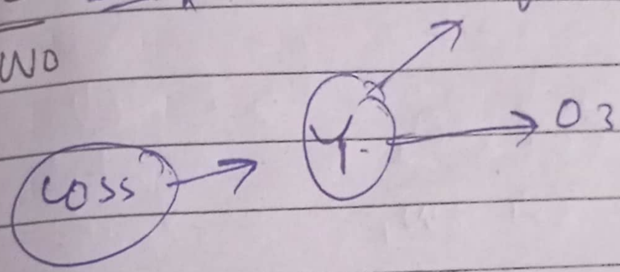
calulate ① $\frac{\delta L}{\delta w_1} = ?$

② $\frac{\delta L}{\delta w_n} = ?$

③ $\frac{\delta L}{\delta w_0} = ?$

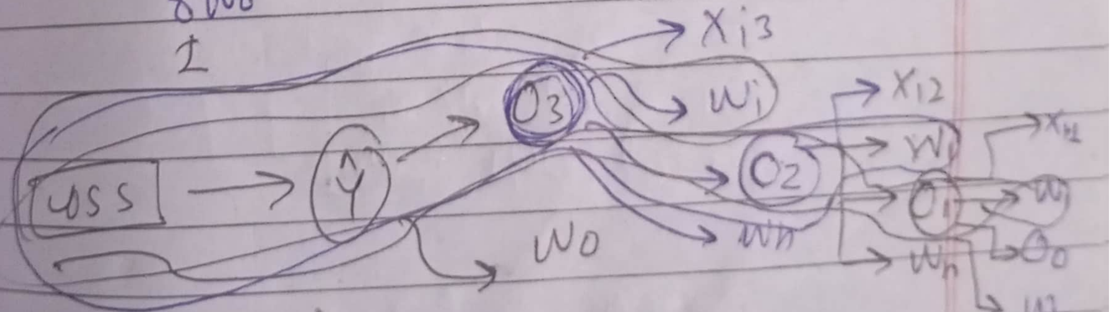
$\frac{\delta L}{\delta w_0}, \frac{\delta L}{\delta w_1}, \frac{\delta L}{\delta w_n}$

$\frac{\delta L}{\delta w_0} = ?$



$$\frac{\delta L}{\delta w_0} = \frac{\Delta \text{loss}}{\Delta Y} \times \frac{\Delta Y}{\Delta w_0}$$

1



$$\frac{\delta L}{\delta w_1} = \frac{\delta L}{\delta \hat{Y}} \times \frac{\delta \hat{Y}}{\delta O_3} \times \frac{\delta O_3}{\delta w_1} + \frac{\delta \text{loss}}{\delta \hat{Y}} \times \frac{\delta \hat{Y}}{\delta O_3} \times \frac{\delta O_3}{\delta O_2} \times \frac{\delta O_2}{\delta w_1}$$

$$+ \frac{\delta \text{loss}}{\delta \hat{Y}} \times \frac{\delta \hat{Y}}{\delta O_3} \times \frac{\delta O_3}{\delta O_2} \times \frac{\delta O_2}{O_1} \times \frac{\delta O_1}{\delta w_1}$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial O_3} \times \frac{\partial O_3}{\partial w_1} +$$

$$\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial O_3} \times \frac{\partial O_3}{\partial O_2} \times \frac{\partial O_2}{\partial w_1} +$$

$$\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial O_3} \times \frac{\partial O_3}{\partial O_2} \times \frac{\partial O_2}{\partial O_1} \times \frac{\partial O_1}{\partial w_i}$$

$$\frac{\partial L}{\partial w_1} = \sum_{j=1}^{3} \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial O_j} \frac{\partial O_j}{\partial w_i}$$

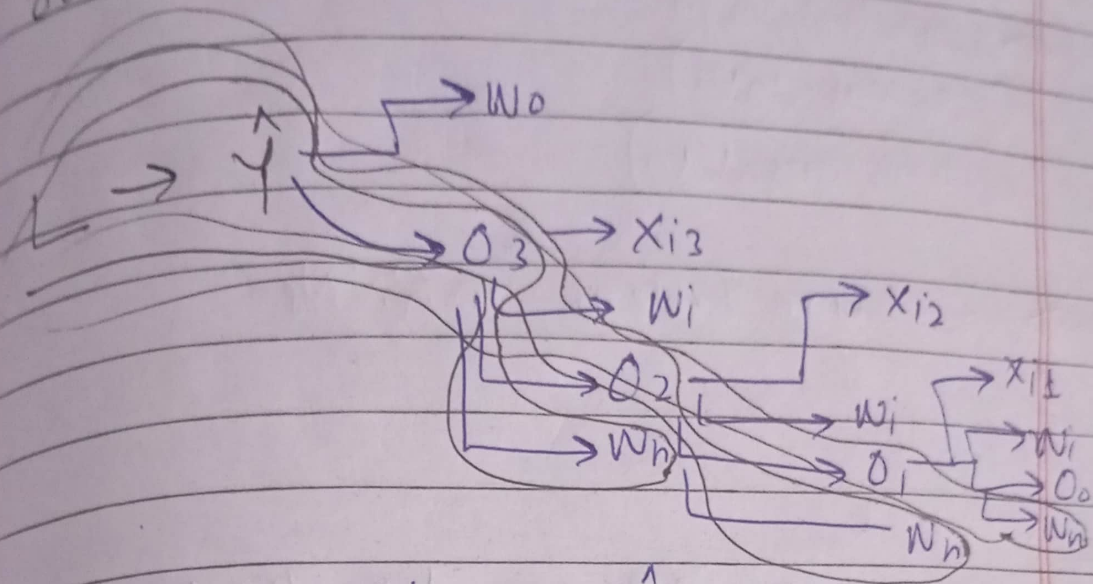
for $j=1$ $\quad \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial O_1} \times \frac{\partial O_1}{\partial w_i}$

$$= \frac{\partial L}{\partial \hat{y}}$$

$$\boxed{\frac{\partial L}{\partial w_i} = \sum_{j=1}^{n} \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial O_j} \times \frac{\partial O_j}{\partial w_i}}$$

n = timesteps

$$\frac{\partial L}{\partial w_h} = ?$$



$$\frac{\partial L}{\partial w_h} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial O_3} \times \frac{\partial O_3}{\partial w_h} +$$

$$\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial O_3} \times \frac{\partial O_3}{\partial O_2} \times \frac{\partial O_2}{\partial w_h} +$$

$$\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial O_3} \times \frac{\partial O_3}{\partial O_2} \times \frac{\partial O_2}{\partial O_1} \times \frac{\partial O_1}{\partial w_h}$$

$$\boxed{\frac{\partial L}{\partial w_h} = \sum_{i=1}^{n} \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial O_j} \frac{\partial O_j}{\partial w_h}}$$

↳ This formula